# STELLAR NUCLEOSYNTHESIS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

*Diagram-based questions on Stellar Nucleosynthesis*

## Question 1

10.0 points

Examine the proton-proton (p-p) chain reaction depicted in the diagram below for main sequence stars. Identify and explain the sequence of nuclear reactions occurring in the p-p chain, including the role of neutrinos and gamma rays in energy transport within the star.

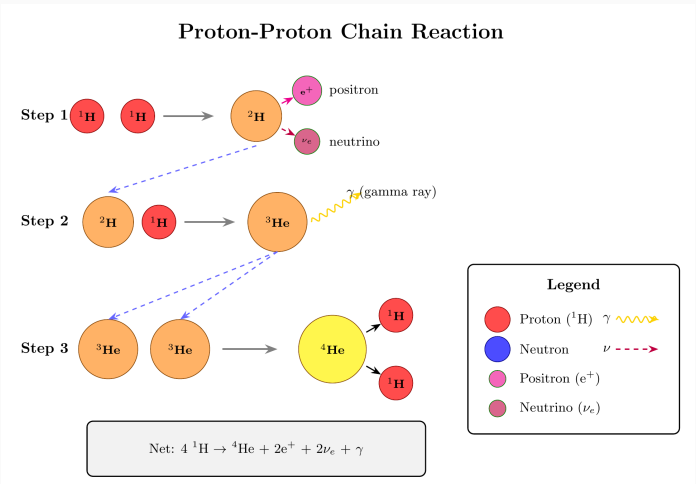


Fig. 1. A diagram illustrating the proton-proton (p-p) chain reaction in main sequence stars. The diagram should show the sequence of nuclear reactions starting with two protons (1H) fusing to form deuterium (2H), releasing a positron and a neutrino. Then, deuterium fuses with another proton to form helium-3 (3He) and emits a gamma ray. Finally, two helium-3 nuclei fuse to form helium-4 (4He), releasing two protons. Include labels for each particle and reaction step, and indicate the emission of neutrinos and gamma rays. IMPORTANT correction needed: The diagram explicitly illustrates the entire sequence of. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

## Question 2

10.0 points

Analyze the CNO cycle reaction network depicted in the diagram below for massive stars. Describe how this cycle contributes to the stellar nucleosynthesis process and compare its efficiency to the proton-proton chain in terms of energy production.

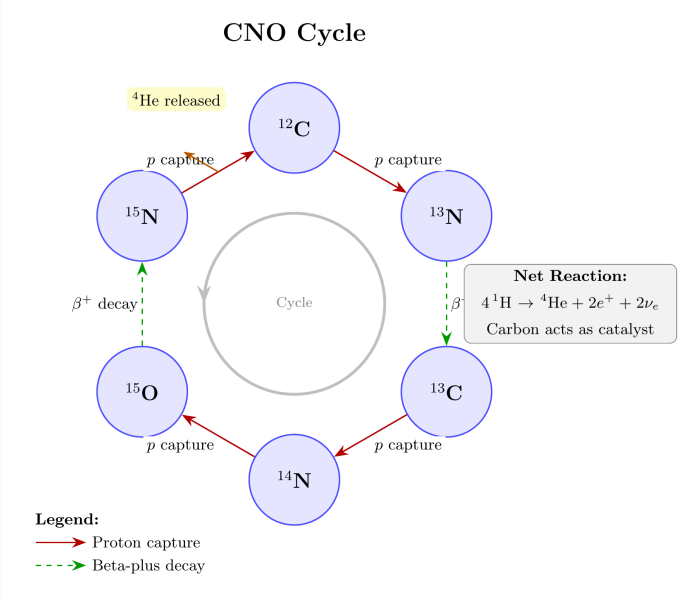


Fig. 2. A diagram of the CNO cycle reaction network for massive stars. Include the main isotopes involved: carbon-12 (C-12), nitrogen-14 (N-14), oxygen-16 (O-16), and their interactions. Show the sequence of reactions starting with C-12, capturing protons, and transforming into N-14 and O-16, eventually returning to C-12. Include arrows to indicate the flow of reactions and label each isotope. Highlight the energy release at each step to compare with the proton-proton chain. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires showing the energy release. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires showing the energy release at. Ensure ALL structural components mentioned in the description are clearly labeled.

## Question 3

10.0 points

Interpret the Hertzsprung-Russell (H-R) diagram with evolutionary tracks shown in the diagram below. Explain how the diagram represents the life cycles of different mass stars and identify the key stages for a one-solar-mass star.

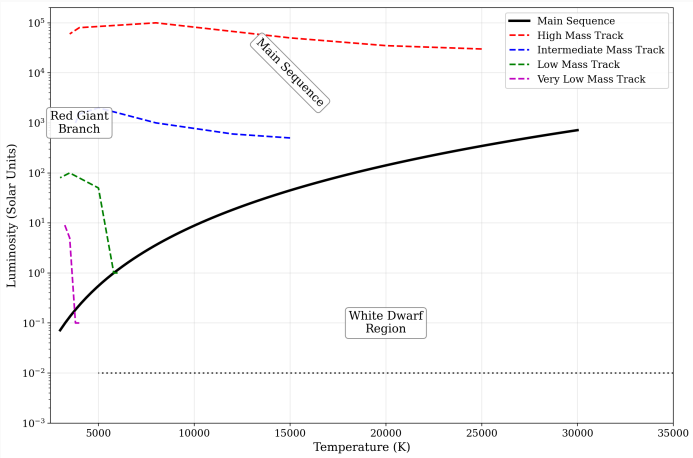


Fig. 3. Hertzsprung-Russell (H-R) diagram showing evolutionary tracks for different mass stars. The diagram plots stellar luminosity (vertical axis) against surface temperature (horizontal axis, decreasing to the right). Include tracks for a one-solar-mass star, highlighting key stages such as main sequence, red giant, and white dwarf phases. Also show tracks for stars of higher and lower masses for comparison. IMPORTANT: The diagram is missing required labels. The diagram description requires highlighting. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires a track for. Ensure ALL structural components mentioned in the description are clearly labeled.

**Question 4** 10.0 points

Examine the binding energy per nucleon curve in the diagram below, focusing on the iron peak. Discuss how this curve illustrates the energy production limits in stellar fusion and why iron signifies an endpoint for nucleosynthesis in stars.

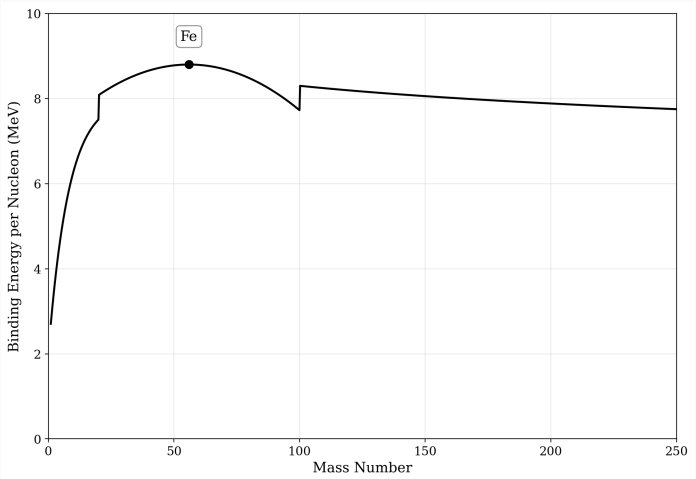


Fig. 4. A binding energy per nucleon curve showing the relationship between the binding energy per nucleon and the atomic mass number. The curve should highlight the iron peak, illustrating the maximum binding energy per nucleon around iron (Fe), and show how the curve slopes downwards for elements heavier than iron, indicating the endpoint for nucleosynthesis in stars.

**Question 5** 10.0 points

Examine the onion-shell model diagram of a pre-supernova massive star's internal structure shown below. Describe the different shells and their associated nuclear processes, explaining how this structure leads to a supernova event.

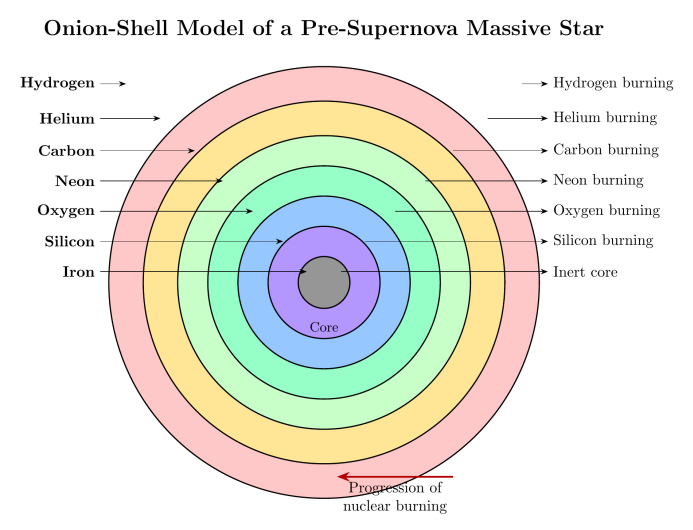


Fig. 5. A diagram of the onion-shell model of a pre-supernova massive star. The diagram should show concentric shells representing different layers of the star. Each shell should be labeled with the nuclear process occurring in that layer, such as hydrogen burning, helium burning, carbon burning, neon burning, oxygen burning, silicon burning, and the iron core at the center. The layers should be arranged from the outermost hydrogen shell to the innermost iron core. Include arrows or labels indicating the progression of nuclear processes from the outer layers to the core. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly labels the nuclear processes,. Regenerate using EXACTLY the values and sequences specified. IMPORTANT correction needed: The diagram explicitly labels all the nuclear. Do NOT include region labels (Feasible/ Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.